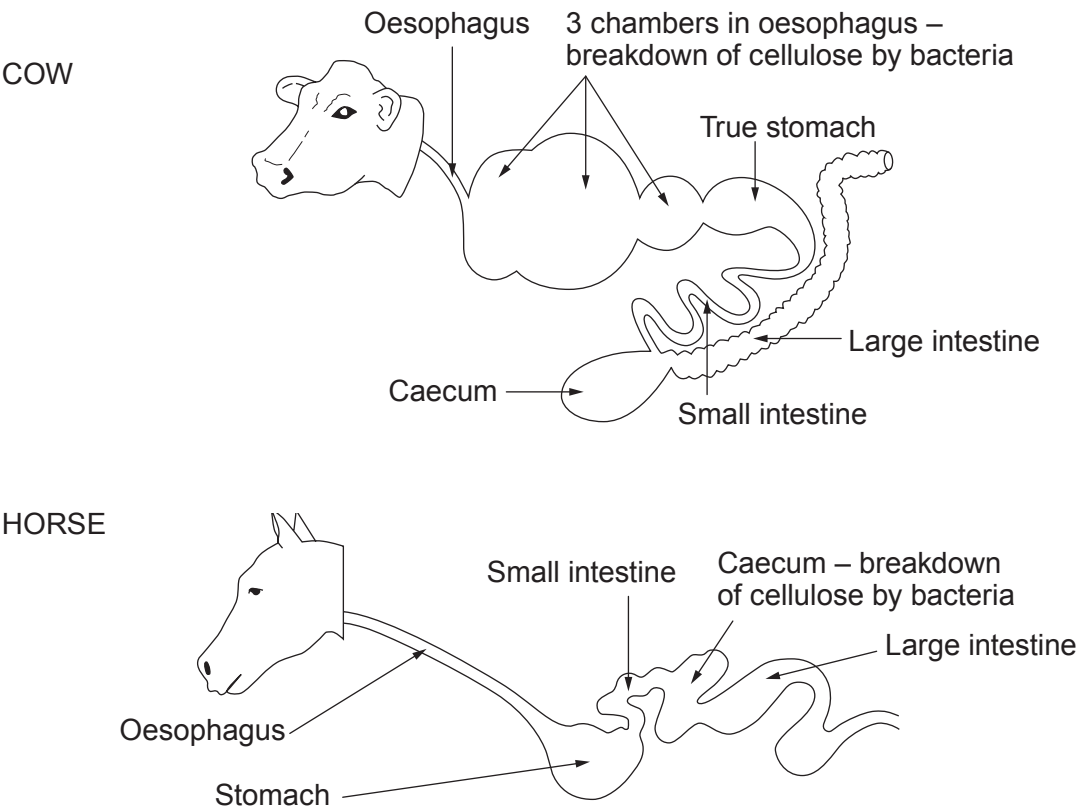


7. All animals need a source of amino acids for growth and repair of tissues. Bacteria, found in the gut of herbivores, can break down cellulose into energy rich molecules; these can be absorbed and used by the herbivore.

Cows produce a large volume of saliva which contains urea. The urea provides a source of nitrogen that bacteria use to make proteins.

Horses need a diet which is much richer in protein than the diet required by a cow.

Manure from horses contains 3 times as much organic nitrogen as cow manure.



Describe how proteins are digested and made available to the muscles of the herbivore. Using the diagrams of the gut of a cow and a horse, account for the difference in the protein requirement of a horse and a cow and the difference of organic nitrogen in the manure of both animals.

[9 QER]

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